

BEYOND MEAN FOILS: AUDITING WORST-FOIL SPECIFICITY IN FROZEN CLIP REGION EXPLANATIONS

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ABSTRACT

A region can overlap the target object and still contribute more to another class. We study this mismatch in frozen CLIP using regions selected by Cluster-based Concept Importance (CCI). Across COCO and VOC with two checkpoints, 41.08–64.78% of regions that pass overlap and mean-contrast checks fail against the strongest competing class. Removing competitors annotated in the image leaves 39.69–63.64% failing. To understand whether region selection can resolve these failures, we examine all eight candidates per image. An alternative passes the test for 6.25–7.84% of failures on COCO and 27.40–31.15% on VOC. Requiring it to preserve the original target-score drop within $\epsilon = .02$ reduces these rates to 0.16–0.98%. The main constraints are therefore the available regions and the target-drop tolerance; relaxing the tolerance increases repair opportunities.

Index Terms— CLIP, explanation auditing, class specificity, constrained region selection

1 INTRODUCTION

Cluster-based Concept Importance (CCI) [2] explains CLIP [1] by grouping patches into regions and measuring how masking them changes image–text scores. We examine CCI-top1, the region whose removal most reduces the target score.

A large target drop does not establish class specificity: removing the same region may reduce another class’s score even more. Comparing with the average competing class can hide this case. We instead test the strongest competitor, or worst foil. We ask how often this test fails and whether another region can pass while preserving the target drop.

Wang and Wang [3] study mean, max, and weighted class contrasts. Their max contrast uses the highest original-image non-target logit; our worst foil has the largest drop after masking a region. COCOA [4] explains corpus similarity using foil samples; CASE [5] constructs class-sensitive saliency. Related work examines CLIP attribution [6, 7, 8], class contrasts [9], and caption foils [10]. We audit a fixed set of images, control for competing objects and foil count, then check all candidate regions to locate constraints on repair.

2 AUDIT AND REPAIR OPPORTUNITY

Figure 1 shows candidate construction (a), selection with one prompt (b), and evaluation with three other prompts (c). The selected region stays fixed during evaluation.

Let x be an image, t its target class, and c any class. Each region R is a set of patches drawn from the candidate set $\mathcal{R}(x)$, which contains $K = 8$ regions. Write $s_m(x, c)$ for checkpoint m ’s pre-softmax image–text score, including CLIP’s learned logit scale. Masking attention to R gives the intervened input $\mathcal{I}(x, R)$ and the score drop

$$d_R(c) = s_m(x, c) - s_m(\mathcal{I}(x, R), c). \quad (1)$$

Positive $d_R(c)$ means a score reduction. Superscript *sel* denotes the prompt “a photo of a {category}”. CCI-top1 selects

$$R_{\text{CCI}} = \arg \max_{R \in \mathcal{R}(x)} d_R^{\text{sel}}(t). \quad (2)$$

Evaluating class specificity. We evaluate the selected region with “a picture of the {category}”, “an image containing a {category}”, and “the {category}”. For these $H = 3$ prompts, $e_h(c)$ is the unit-norm text embedding and $d_{R,h}(c)$ is the score drop in Eq. 1. We average the text embeddings and normalize the result. Equivalently, with $n_c = \|H^{-1} \sum_h e_h(c)\|_2$, the aggregate drop is $\bar{d}_c(R) = H^{-1} \sum_h d_{R,h}(c)/n_c$.

A foil is a non-target class, and $\mathcal{F}_t$ is the set of all such classes. The worst-foil margin compares the target drop with the largest foil drop:

$$P(R, t) = \bar{d}_t - \max_{f \in \mathcal{F}_t} \bar{d}_f. \quad (3)$$

With R implicit, $P < 0$ means a foil loses more score than the target. We call this a failure; a transition to $P \geq 0$ is a sign repair. Normalization can change class rankings, so we also test raw drops in Table 4. Target-response tables use the raw average $H^{-1} \sum_h d_{R,h}(t)$.

We focus on regions that already pass two familiar checks. At least half of their patch centers must lie inside target-class boxes (bbox precision $\geq .5$), and their target drop must exceed the mean foil drop. The images passing both form the fixed baseline set $A = \{\text{bbox} \geq .5, \bar{d}_t > \text{mean}_{f \in \mathcal{F}_t} \bar{d}_f\}$.

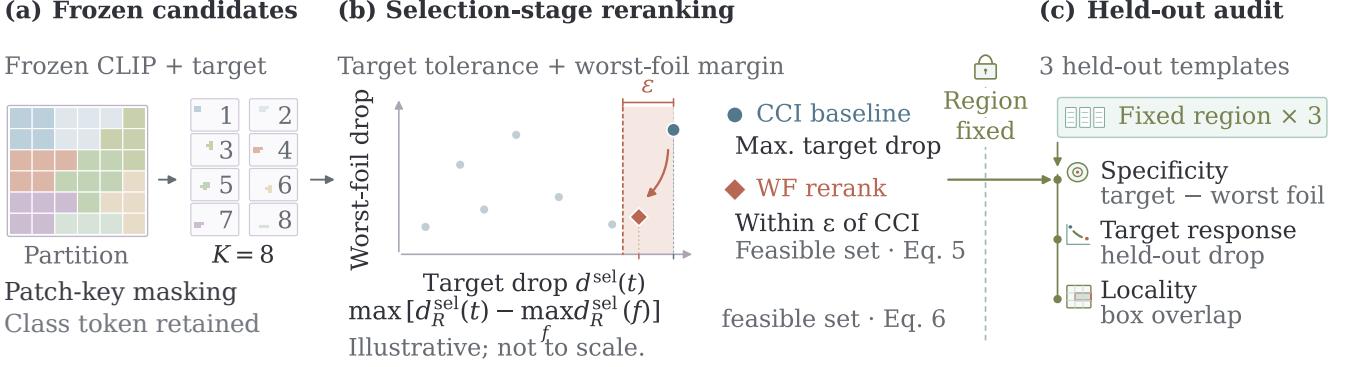


Figure 1: Selection and held-out audit. A region is fixed using selection-stage responses before three-template evaluation of specificity, target response, and locality. Schematic responses illustrate the selection rule.

The failure set is $F = \{i : P_i < 0\}$. Its screened subset is $B = A \cap F$, where i indexes images. We keep both sets fixed when comparing regions. To check whether the target contribution itself is positive, we also report results on $A^+ = A \cap \{\bar{d}_t > 0\}$.

Selecting an alternative region. We allow the target drop to decrease by at most $\epsilon \geq 0$ relative to CCI-top1. Within this tolerance, worst-foil (WF) reranking selects the region with the largest selection-prompt margin. Figure 1(b) illustrates the choice. The margin and feasible set are

$$M^{\text{sel}}(R, t) = d_R^{\text{sel}}(t) - \max_{f \in \mathcal{F}_i} d_R^{\text{sel}}(f), \quad (4)$$

$$\mathcal{R}_\epsilon = \{R : d_R^{\text{sel}}(t) \geq d_{R_{\text{CCI}}}^{\text{sel}}(t) - \epsilon\}, \quad (5)$$

and the selected region is

$$R_{\text{WF}} = \arg \max_{R \in \mathcal{R}_\epsilon} M^{\text{sel}}(R, t). \quad (6)$$

CCI is feasible by construction. We use $\epsilon = .02$ logit units and sweep .01/.02/.05/.10/.20 within each checkpoint.

Locating the constraints on repair. For each image i , write $P_i(R) = P(R, t_i)$ for its target t_i . To find out whether repair is possible, an oracle checks all candidates using their held-out margins. With $\mathbf{1}\{\cdot\}$ the indicator (one if true, zero otherwise), the feasible sign-repair count on fixed baseline B is

$$N_O(B, \epsilon) = \sum_{i \in B} \mathbf{1}\left\{ \max_{R \in \mathcal{R}_\epsilon(i)} P_i(R) \geq 0 \right\}. \quad (7)$$

The unrestricted oracle removes the budget constraint and searches all K candidates. Each image in B falls into one of four groups: no candidate has $P \geq 0$ (C_0); a candidate passes but none meets the budget (C_1); a feasible candidate passes but WF misses it (C_2); or WF repairs the failure (r). The counts satisfy $|B| = C_0 + C_1 + C_2 + r$ and $N_O = C_2 + r$. A sign repair need not preserve overlap or positive target contribution. We also test the joint condition J : the new region must have

$P \geq 0$, pass the overlap and mean-foil checks, and have $\bar{d}_t > 0$. We count these repairs as j , keeping baseline B fixed. Unlike the oracle, WF selects using only the selection prompt.

Controlling for competing objects. A foil may name another object in the image. To measure its contribution, remove all annotated-present classes from image i 's foil set, leaving $\mathcal{F}_i^{\text{abs}}$. The protocol holds R_{CCI} and full-foil screen A fixed; F_i^{abs} tests $\bar{d}_t - \max_{f \in \mathcal{F}_i^{\text{abs}}} \bar{d}_f < 0$. Removing any foils can reduce failure, so we compare this exclusion with randomly retaining the same number of foils. For N_i full foils, $k_i = |\mathcal{F}_i^{\text{abs}}|$, and m_i foils with $\bar{d}_f > \bar{d}_t$, a uniform size- k_i subset without replacement fails with probability

$$q_i = 1 - \frac{\binom{N_i - m_i}{k_i}}{\binom{N_i}{k_i}}. \quad (8)$$

The numerator is zero when $k_i > N_i - m_i$. The paired difference is $\Delta_A = |A|^{-1} \sum_{i \in A} (\mathbf{1}\{F_i^{\text{abs}}\} - q_i)$. Negative Δ_A favors annotation-based over random removal. Retained foil sets are nonempty.

3 PROTOCOL AND REPRODUCIBILITY

COCO val2014 [11] and VOC2007 test [12] provide 27,708/1,943 eligible images, each evaluated with OpenAI CLIP B/16 and B/32 (ViT-B with 16- or 32-pixel patches; 59,302 image-model records). The target is the largest instance by annotation area on COCO or box area on VOC. Images require another category and 2–80% target area; COCO crowd annotations are excluded. Full foils are the other 79/19 classes.

K-means clusters L2-normalized projected patch features: $K = 8$, three initializations, 50 iterations, seed 1701 + i (sorted IDs shuffled with seed 1701 define i). Intervention $\mathcal{I}$ adds $-\infty$ to attention logits at selected patch-key columns before softmax in every block/head, retaining the class-token column. Ties use first argmax. Target boxes are mapped by $224 / \min(W_{\text{img}}, H_{\text{img}})$ and center-crop offsets, where $W_{\text{img}}, H_{\text{img}}$ are the image width and height [13].

4 RESULTS AND ANALYSIS

Table 1: Fixed baseline A : full-foil, annotation-absent, and exact cardinality-matched random failure rates (%). Δ_A is absent minus random in percentage points, with paired 95% CI.

Setting	Full	Absent	Random	Δ_A
COCO B/16	64.37	63.17	64.09	-0.92 [-1.08,-0.77]
COCO B/32	64.78	63.64	64.50	-0.86 [-1.01,-0.71]
VOC B/16	42.27	40.94	41.46	-0.52 [-1.16,+0.06]
VOC B/32	41.08	39.69	40.24	-0.55 [-1.21,+0.04]

Removing annotated competitors leaves most failures. Table 1 compares the same baseline images before and after removing classes annotated in the image. The set A contains 17,726, 17,848, 1,209, and 1,227 images in table order. Failure falls by only 1.14–1.39 percentage points: 96.63–98.24% of the failures in B remain. On COCO, exclusion helps more than removing the same number of classes at random. On VOC, the confidence intervals for that difference include zero. Annotated competing objects thus explain only a small part of the observed failure.

The pattern also holds when the target contribution is positive. In A^+ , failure after exclusion remains 61.38–62.62% on COCO and 39.26–40.05% on VOC. Even COCO images with exactly two annotated categories have failure rates of 63.39% and 64.95% for B/16 and B/32, respectively. The mismatch therefore persists beyond crowded scenes.

Table 2: Exact local decomposition of *screened failures* $B = A \cap F$ at $\epsilon = .02$. Counts in C_0, C_1, C_2, r are mutually exclusive; $C_2 + r$ is the feasible-oracle count.

Setting	$ B $	C_0	C_1	C_2	r
COCO B/16	11410	10697	695	0	18
COCO B/32	11562	10655	876	1	30
VOC B/16	511	371	135	0	5
VOC B/32	504	347	154	1	2

Few failures have an alternative within the budget. Table 2 shows where repair stops. For 92.16–93.75% of COCO failures and 68.85–72.60% of VOC failures, none of the eight candidates passes the worst-foil test (C_0). Figure 2 shows the remaining opportunity as blue circles: 6.25–7.84% on COCO and 27.40–31.15% on VOC. Imposing $\epsilon = .02$ removes 96.43–98.09% of these opportunities, leaving only 0.16–0.98% of B repairable (orange diamonds). WF misses at most one of these feasible repairs per setting (C_2). The main losses occur before the final ranking decision.

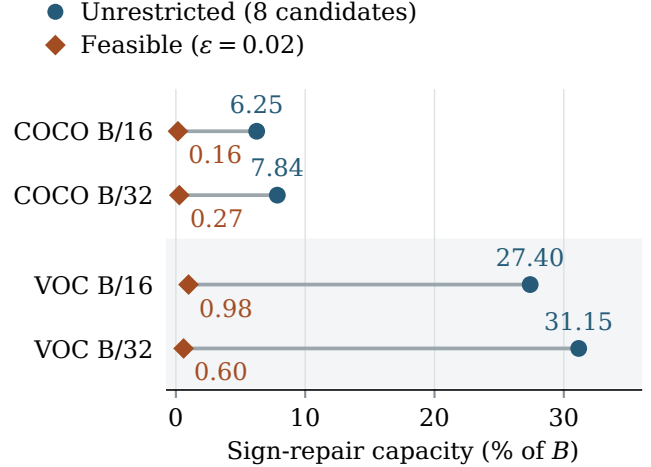


Figure 2: Budget-constrained sign-repair capacity among fixed screened failures B . Connected markers compare the same setting on a shared linear scale. Unrestricted: at least one of all eight candidates has $P \geq 0$; feasible: at least one also meets $\epsilon = .02$. Percentages use each setting’s own $|B|$ (Table 2) and are computed by exact enumeration.

Table 3: Budget endpoints (C: COCO; V: VOC). Multi: percentage of all eligible images with multiple feasible candidates. O_B, r_B : oracle/actual sign-repair percentages of screened failures B . $\Delta d_t, \Delta b$: all-image raw-target and bbox changes (bbox in percentage points).

Set	ϵ	Multi	O_B	r_B	Δd_t	Δb
C16	0.02	4.71	0.16	0.16	+0.0001	-0.12
C16	0.20	35.78	1.57	1.50	-0.0140	-1.63
C32	0.02	4.40	0.27	0.26	-0.0001	-0.08
C32	0.20	34.95	2.05	2.00	-0.0143	-0.60
V16	0.02	4.79	0.98	0.98	-0.0008	-0.34
V16	0.20	34.48	6.65	6.07	-0.0223	+0.08
V32	0.02	4.73	0.60	0.40	-0.0007	+0.02
V32	0.20	34.64	7.14	6.75	-0.0161	+1.07

A larger tolerance opens up more alternatives. At $\epsilon = .02$, about 95% of images have only one feasible candidate (Table 3). Increasing the tolerance to .20 gives multiple choices on about 35% of images. Oracle repair capacity rises to 1.57–2.05% of COCO failures and 6.65–7.14% of VOC failures. Here $O_B = 100N_O/|B|$ is the percentage that an oracle can repair, and $r_B = 100r/|B|$ is the percentage actually repaired by WF.

The extra choice changes what the selected region captures. The table’s Δd_t and Δb are WF minus CCI changes averaged over all eligible images; bbox changes are in percentage points. At .20, raw target drops decrease in all four settings, while overlap decreases on COCO and increases on VOC. More repair opportunities therefore do not imply a uniform improvement in target response or localization.

Table 4: Normalization sensitivity on fixed earlier-run A_{old} and earlier-run CCI regions. Raw/Norm: failure rates (%); Flip: percentage whose failure status differs. A_{old} denotes earlier-run screen-passers.

Setting	$ A_{\text{old}} $	Raw	Norm	Flip
COCO B/16	17,726	64.318	64.369	0.107
COCO B/32	17,847	64.762	64.778	0.073
VOC B/16	1,209	42.184	42.266	0.083
VOC B/32	1,227	41.157	41.157	0.000

Normalization rarely changes the outcome. Table 4 compares raw and normalized scores while holding the earlier-run regions and images fixed. Failure rates differ by at most 0.083 percentage points, and at most 0.107% of images switch between passing and failing. The worst foil is unchanged on 99.42–99.84% of images. For example, COCO B/16 moves from 64.318% to 64.369% failure, while VOC B/32 stays at 41.157%. The high failure rate is therefore present under both score definitions.

Table 5: Actual sign repairs *within fixed baseline B*, $\epsilon = .02$. r counts repairs; j counts those also satisfying J . WF–CCI means with repair-subset 95% CIs. Δd_t is raw held-out target-score change; only bbox change is in percentage points.

Setting	r	j	Δd_t	100 Δbbox
COCO B/16	18	12	0.029	-19.53
			[-0.096,0.148]	[-38.12,-1.32]
COCO B/32	30	25	0.094	-14.04
			[0.036,0.148]	[-27.70,-1.74]
VOC B/16	5	3	-0.021	-16.87
			[-0.170,0.182]	[-26.11,-7.00]
VOC B/32	2	1	-0.078	-28.17
			[-0.115,-0.042]	[-78.57,22.22]

Repairing specificity can reduce spatial overlap. Table 5 examines only the failures that WF repairs, whereas Table 3 averages over all eligible images. Among these repairs, mean bbox precision falls by 14.04–28.17 percentage points. Mean target-score changes are positive on COCO and negative on VOC. Passing the worst-foil test therefore does not guarantee that the new region retains the original region’s overlap or target response.

The counts r and j make this distinction explicit. On COCO B/32, WF repairs 30 failures, but only 25 also satisfy the overlap, mean-foil, and positive-target checks in J . The same inequality $j < r$ holds in every setting. Even the feasible oracle finds only 12, 26, 3, and 2 repairs satisfying all checks in table order. Changes can also go the other way: among all baseline passes on COCO B/32, four become worst-foil failures. These regressions are outside the fixed failure set B .

Table 6: Nonzero-threshold sensitivity on fixed baseline A . Entries are percentages of screen-passers A with $P < -\delta$, where $\delta \geq 0$ is the required failure magnitude in the normalized margin. Thresholds are evaluated post hoc in logit units.

Setting	$\delta = 0$	.01	.10	.50
COCO B/16	64.37	63.94	59.49	36.05
COCO B/32	64.78	64.38	60.22	35.19
VOC B/16	42.27	41.44	36.23	13.73
VOC B/32	41.08	40.26	34.39	11.90

The failures are not confined to margins near zero. Table 6 tightens the test from $P < 0$ to $P < -\delta$. At $\delta = .10$, 59.49–60.22% of COCO screen-passers and 34.39–36.23% of VOC screen-passers still fail. Rates decline further at .50. Thus many failures survive a stricter margin requirement, although their prevalence depends on the chosen threshold.

Implications for region selection. The comparisons point to two distinct changes. Table 2 identifies images that need a better candidate region (C_0) and images that already have a passing region outside the budget (C_1). Figure 2 shows how sharply the budget narrows the available choices, and Table 3 shows the effect of relaxing it. Changing the ranking rule alone can recover only the additional feasible opportunities in C_2 . Improving repair therefore requires attention to which regions are proposed and how much target-drop loss is allowed.

Reproducibility and uncertainty. The primary tables come from one run that saved candidate masks, selection and evaluation responses, and bbox records together. Validation checks reconstruct the regions, scores, choices, and failure counts. The released records include candidate summaries, checks, configuration, and hashes; masks and full responses remain local. Table 4 uses earlier-run records. Three images changed selected regions between runs.

We obtain confidence intervals from 10,000 image-level bootstrap samples, using the 2.5th and 97.5th percentiles without multiplicity correction. Paired effects are differenced within each image. Foil comparisons use fixed A or A^+ ; repair-cost intervals use only actual repairs. We use exact q_i values, checked against 100 seeded random draws.

5 CONCLUSION

A region can overlap the target and beat the average competing class while contributing more to a single competitor. This occurs frequently for CCI-top1, even after annotated competitors are removed. Checking every candidate shows why reranking repairs so few failures: many images have no passing alternative, and most remaining alternatives exceed the target-drop tolerance. Relaxing that tolerance makes more repairs possible but also changes target response and overlap. Worst-foil comparisons therefore add information that overlap and mean contrast alone do not provide when evaluating frozen CLIP region explanations.

REFERENCES

- [1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, “Learning transferable visual models from natural language supervision,” in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139. PMLR, 2021, pp. 8748–8763. [Online]. Available: <https://proceedings.mlr.press/v139/radford21a.html>.
- [2] A. Agarwal, S. Karanam, and V. Gandhi, “Concept regions matter: Benchmarking CLIP with a new cluster-importance approach,” *arXiv preprint arXiv:2511.12978*, 2025. [Online]. Available: <https://arxiv.org/abs/2511.12978>.
- [3] Y. Wang and X. Wang, “‘Why not other classes?’: Towards class-contrastive back-propagation explanations,” in *Advances in Neural Information Processing Systems*, vol. 35, 2022. [Online]. Available: <https://doi.org/10.52202/068431-0660>.
- [4] C. Lin, H. Chen, C. Kim, and S.-I. Lee, “Contrastive corpus attribution for explaining representations,” in *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: <https://arxiv.org/abs/2210.00107>.
- [5] D. Williamson, Y. Ji, and M. Dwyer, “CASE: Contrastive activation for saliency estimation,” *arXiv preprint arXiv:2506.07327*, 2025. [Online]. Available: <https://arxiv.org/abs/2506.07327>.
- [6] Y. Wang, T. G. J. Rudner, and A. G. Wilson, “Visual explanations of image-text representations via multi-modal information bottleneck attribution,” in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 16 009–16 027. [Online]. Available: <https://doi.org/10.52202/075280-0704>.
- [7] Y. Li, H. Wang, Y. Duan, J. Zhang, and X. Li, “A closer look at the explainability of contrastive language-image pre-training,” *Pattern Recognition*, vol. 162, p. 111409, 2025. [Online]. Available: <https://doi.org/10.1016/j.patcog.2025.111409>.
- [8] M. Dreyer, L. Hufe, J. Berend, T. Wiegand, S. Lapuschkin, and W. Samek, “From what to how: Attributing CLIP’s latent components reveals unexpected semantic reliance,” *arXiv preprint arXiv:2505.20229*, 2025. [Online]. Available: <https://arxiv.org/abs/2505.20229>.
- [9] F. Li, Z. Meng, C. Tan, and Y. Yin, “Class-disentangled attribution with gradient guidance for CLIP,” *Journal of King Saud University Computer and Information Sciences*, vol. 38, p. 376, 2026. [Online]. Available: <https://doi.org/10.1007/s44443-026-00779-3>.
- [10] M. Yuksekgonul, F. Bianchi, P. Kalluri, D. Jurafsky, and J. Zou, “When and why vision-language models behave like bags-of-words, and what to do about it?” in *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: <https://openreview.net/forum?id=KRLUvvh8uaX>.
- [11] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollar, and C. L. Zitnick, “Microsoft COCO: Common objects in context,” in *Computer Vision—ECCV 2014*, ser. Lecture Notes in Computer Science, vol. 8693. Springer, 2014, pp. 740–755. [Online]. Available: https://doi.org/10.1007/978-3-319-10602-1_48.
- [12] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, “The PASCAL visual object classes (VOC) challenge,” *International Journal of Computer Vision*, vol. 88, no. 2, pp. 303–338, 2010. [Online]. Available: <https://doi.org/10.1007/s11263-009-0275-4>.
- [13] T. Gomez, T. Freour, and H. Mouchere, “Metrics for saliency map evaluation of deep learning explanation methods,” in *Pattern Recognition and Artificial Intelligence: Third International Conference, ICPRAI 2022*, ser. Lecture Notes in Computer Science. Springer, 2022, pp. 84–95. [Online]. Available: https://doi.org/10.1007/978-3-031-09037-0_8.

COMPLIANCE WITH ETHICAL STANDARDS

This work analyses publicly released COCO and PASCAL VOC images and annotations. No participants were recruited and no new human or animal experiments were conducted.